



**R. C. PATEL
INSTITUTE OF TECHNOLOGY**
An Autonomous Institute

Case Study

Advanced Statistics Laboratory (RCP23DCPC602)

Submitted by

- 1. Solunke Rohit Dilip (231106022)**
- 2. Jagtap Divya Sanjay (231106023)**
- 3. Patil Dhiraj Suresh (231106024)**
- 4. Patil Keshar Kailas (231106025)**
- 5. Chavhan Dhanshri Prakash (231106026)**

Department of Computer Science & Engineering (Data Science)



R C Patel Institute of Technology, Shirpur

AY: 2025-26

Vision of the Institute

To become a leading Institute in Technical education fostering innovation, research, ethical values, and sustainable development for the betterment of society.

Mission of the Institute

To impart high quality Technical Education through:

M1. Innovative and Interactive learning process and high quality, globally recognized instructional programs.

M2. Fostering a collaborative scientific temper among students with ethical responsibility towards the society.

M3. Preparing students from diverse backgrounds to have aptitude for employment, entrepreneurship and research with a spirit of Professionalism.

M4. To contribute to nation's sustainable development.

Vision of the Department

To provide cutting-edge Computer Engineering education in Data Science while instilling socio-moral values.

Mission of the Department

M1: To deliver state-of-the-art, ICT-enabled teaching and learning to achieve excellence in Data Science education.

M2: To develop professionally competent Data Science Engineers, meeting evolving industrial and societal needs.

M3: To prepare employable professionals with ethical values and a commitment to professional and social responsibility.

Program Educational Objectives (PEOs) of the Department

PEO 1: Graduates will have proficiency in Artificial Intelligence and Machine Learning, with lifelong learning skills to advance their careers as professionals, entrepreneurs, and leaders.

PEO 2: Graduates will possess strong expertise in Artificial Intelligence and Machine Learning, coupled with lifelong learning abilities to excel as professionals, entrepreneurs, and leaders.

PEO 3: Graduates will apply AI and ML knowledge to solve real-world problems, contributing to sustainable development and societal benefit.

Program Specific Outcomes (PSOs)

PSO 1: Apply programming concepts, algorithms, and data structures to develop data-driven software and web solutions.

PSO 2: Develop intelligent solutions using machine learning, data analysis, and cloud technologies for practical problem-solving.

Program Outcomes:

Engineering graduates will be able to:

PO1: Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2: Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3: Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4: Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5: Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.

PO6: The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7: Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO8: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9: Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10: Communication: Communicate effectively on complex engineering activities with the Engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11: Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO12: Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Advanced Statistics Laboratory (RCP23DLPC602)

CO	Course Outcomes	Blooms Level	Blooms Description
CO1	Interpret and evaluate estimators under classical and Bayesian paradigms.	L2	Understand
CO2	Apply and interpret hypothesis tests for linear and logistic regression models.	L3	Apply
CO3	Apply nonparametric testing methods when parametric assumptions fail.	L3	Apply
CO4	Examine model performance and generalization.	L3	Apply
CO5	Use statistical reasoning with computational implementation.	L3	Apply